



Emerging Trends in Enhancing Model Performance

CSIT205 Generative AI – 2025



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Wollongong

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*SCHOOL OF COMPUTING AND INFORMATION
TECHNOLOGY*

*FACULTY OF ENGINEERING AND INFORMATION
SCIENCES*



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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things. The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi
Artist Lauren Henry.

The background is a vibrant rainbow with horizontal bands of red, orange, yellow, green, cyan, blue, and purple. Large, dark blue letters 'U', 'O', and 'W' are positioned on the right side of the image, with the 'U' at the top, 'O' in the middle, and 'W' at the bottom.

Everyone is welcome here.

UOW takes pride in
the diversity of our
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Subject introduction



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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

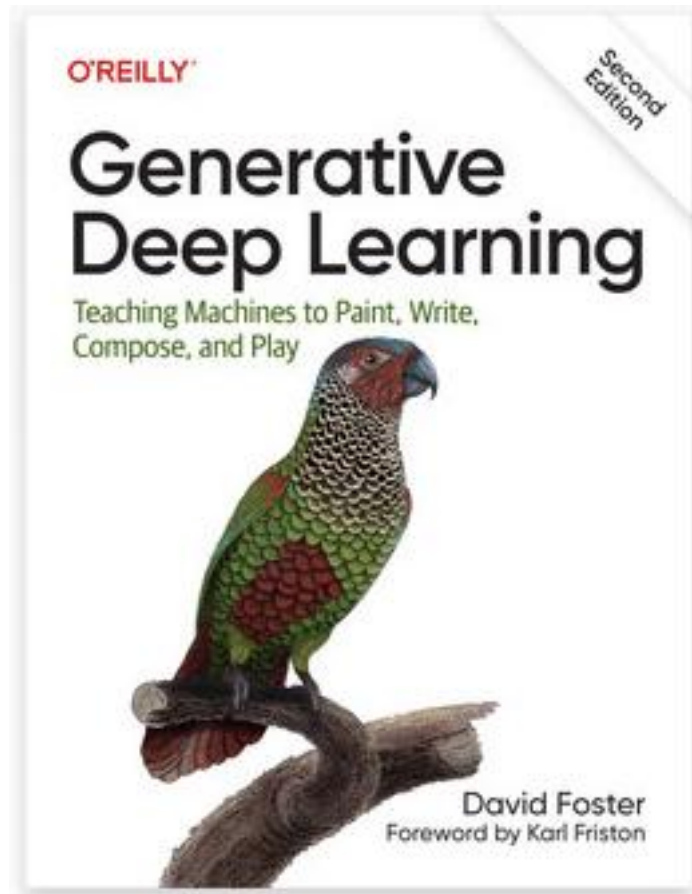
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don't forget 

This is the book we're using



<https://www.oreilly.com/library/view/generative-deep-learning/978109813>

What did we discuss last week?



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Parameter Efficient Fine-Tuning (PEFT)

- *Collection of measures to speed up training:*

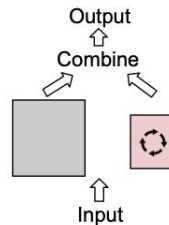
- *Additive Fine-tuning*
- *Selective Fine-tuning*
- *Reparameterised Fine-tuning*
- *Hybrid Fine-tuning*

- *Key mechanism: parts of the model remain 'frozen' (grey), and only a small portion of parameters is updated (red)*

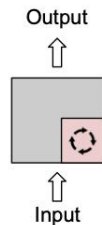
→ *Faster training*

→ *Prevents catastrophic forgetting*

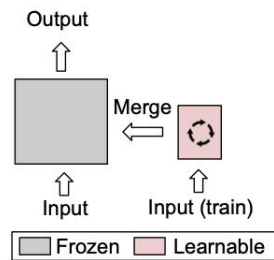
Additive PEFT



Selective PEFT



Reparameterised PEFT



Han et al., 2024

Low Rank Adaptation (LoRA) (Hu *et al.*, 2021)

- Motivation: there exists an 'intrinsic lower dimension', rank r (Aghajanyan *et al.*, 2020) that is just as effective as retraining all parameters for a given task.

- 'Bottleneck adapter', because decomposes the matrix while maintaining as much information as possible.

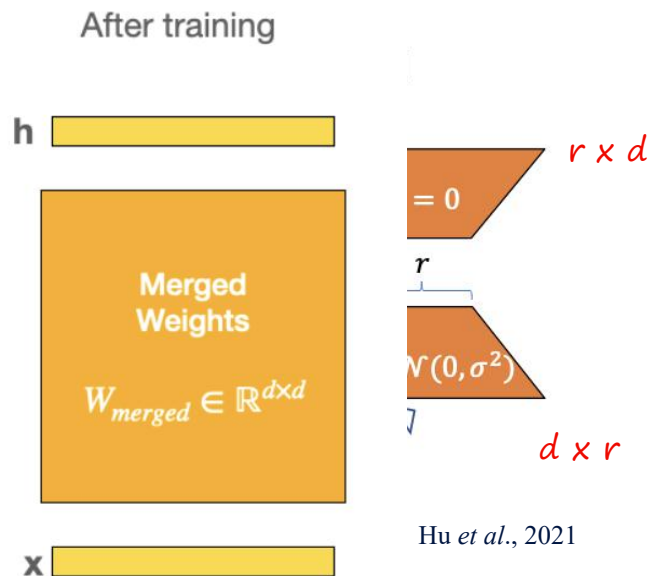
→ All pre-trained model parameters are frozen!

- Training Process:

- Determine a rank r (usually small, like 1 or 2)
- Learn matrices A ($d \times r$) and B ($r \times d$)

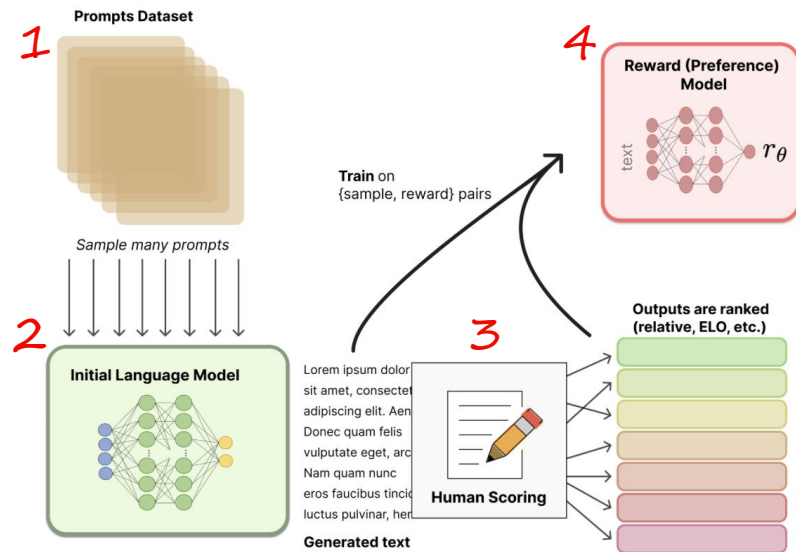
- Merge $A * B$ with pre-trained weights

If $d=64$, and $r=2$, then n learned parameters is $64 * 2 + 2 * 64 = 256$, instead of $64 * 64 = 4096$



Reinforcement Learning with Human Feedback (RLHF)

- Quality of LLMs is subject to human preferences → human feedback determines the reward
- Process:
 1. Collect and present prompts.
 2. Collect multiple LLM responses per prompt.
 3. Human provides a ranking (reward).
 4. A reward model (Stiennon et al, 2022) is trained to calculate the reward for each LLM response.
 5. A policy (not shown) is used to train the initial LM using RL with the reward model providing feedback.



Source: <https://huggingface.co/blog/rlhf>

Week 8: Assessment 1 due Friday 2 May 23.30pm

- *Topic: Prompt engineering*
- *Don't forget to report on your process.*
- *You can now re-submit your assignment.*

 Note: report goes to report box; prompts go to prompt box!

Assignment - Turnitin
Assessment 1 Submission box

My Submissions

Part 1
80%

Part 2
20%

Prompt-answer pairs	Report												
<table><thead><tr><th>Title</th><th>Start Date</th><th>Due Date</th><th>Post Date</th><th>Marks Available</th></tr></thead><tbody><tr><td>Assessment 1 Submission box - Prompt-answer pairs</td><td>15 Apr 2025 - 11:00</td><td>2 May 2025 - 23:30</td><td>19 May 2025 - 23:30</td><td>80 </td></tr></tbody></table>		Title	Start Date	Due Date	Post Date	Marks Available	Assessment 1 Submission box - Prompt-answer pairs	15 Apr 2025 - 11:00	2 May 2025 - 23:30	19 May 2025 - 23:30	80 		
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<table><thead><tr><th>Submission Title</th><th>Turnitin Paper ID</th><th>Submitted</th><th>Similarity</th><th>Grade</th><th>Overall Grade</th></tr></thead><tbody><tr><td> View Digital Receipt test</td><td>2579103587</td><td>4/02/25, 13:24</td><td>0% </td><td>--</td><td>--</td></tr></tbody></table>		Submission Title	Turnitin Paper ID	Submitted	Similarity	Grade	Overall Grade	 View Digital Receipt test	2579103587	4/02/25, 13:24	0% 	--	--
Submission Title	Turnitin Paper ID	Submitted	Similarity	Grade	Overall Grade								
 View Digital Receipt test	2579103587	4/02/25, 13:24	0% 	--	--								
Submit Paper  													

- Upload your report in the Prompt-answer pairs submission box → max. 20/80 marks.
- Upload your prompt in Report submission box → max. 20/20 marks instead of max. 80.

Actions for Lab C

1. Form a group of 4-6 within your Lab C group. Note: max. 5 groups per lab.



Group self-selection

Tue 10.30 - 12.30 - Liverpool CLC/01 



Not available unless: You belong to **AUTM-CSIT205-LP-OC-CLC/01 - 2025,Liverpool,On Campus,Autumn,CSIT205,Class 1**

2. Schedule a 'Chat with the CTO' in week 10.



Scheduler

Chat with the CTO - Tue 10.30-12.30 Liverpool CLC/01 



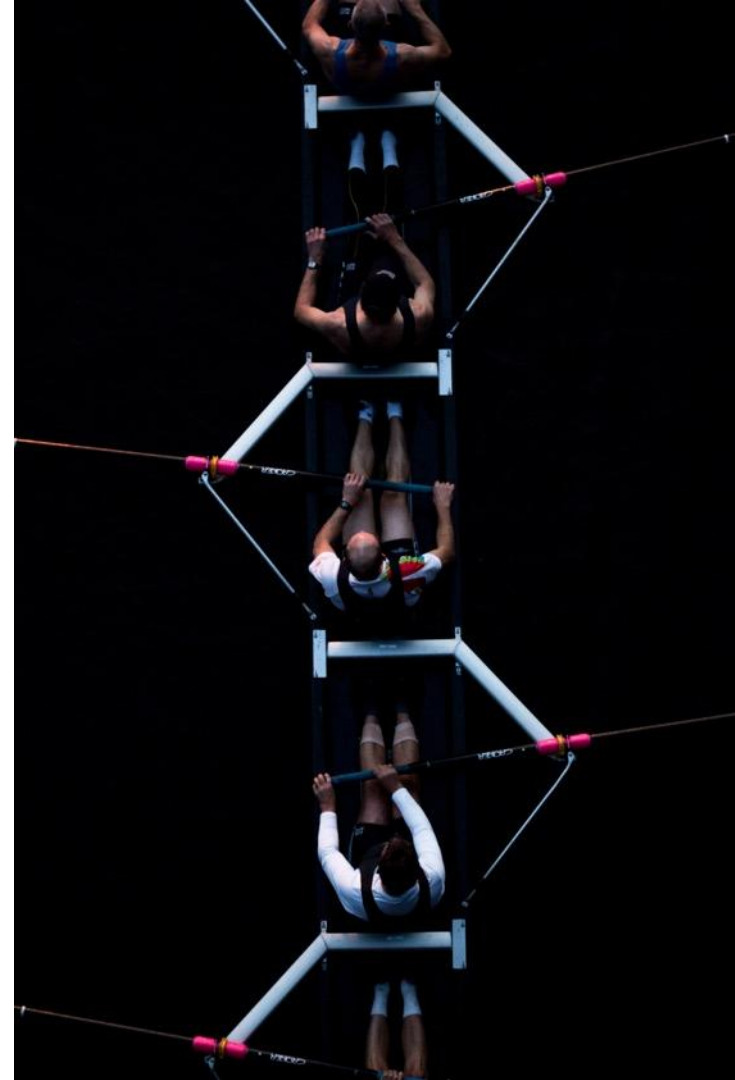
Not available unless: You belong to **AUTM-CSIT205-LP-OC-CLC/01 - 2025,Liverpool,On Campus,Autumn,CSIT205,Class 1**

Why are we here?

CSIT205 Generative AI - Module 2

Emerging Trends in Model Enhancement

- Prompt engineering
- RAG
- Fine-tuning



Prompt engineering



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ReAct (Reason Act) (Yao *et al.*, 2023)

Adds Acts and Observations on top of CoT prompting

- Thoughts, Acts, and Observations show the LLM how thoughts and actions lead to observations (output).
 - Subsequent TAO triples provide feedback and feedforward mechanisms that the LLM can learn from.
- 🏆 Seems to eliminate hallucinations from CoT prompting.
- ✗ Heavily relies on search for additional knowledge.

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: `Search[Apple Remote]`

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...

Thought 2: Apple Remote was originally designed to control the `Front Row media center` program. I need to search `Front Row` next and find what other device can control it.

Act 2: `Search[Front Row]`

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', ' `Front Row (software)` ', ...]

Thought 3: `Front Row` is not found. I need to search `Front Row (software)` .

Act 3: `Search[Front Row (software)]`

Obs 3: `Front Row` is a discontinued media center software ...

Thought 4: `Front Row (software)` is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: `Finish[keyboard function keys]`



Directional Stimulus Prompting (Li et al., 2023)

Adds hints to the prompt to provide direction

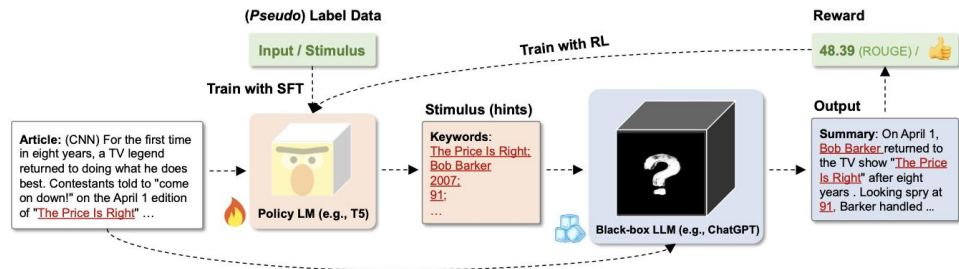
- Hints generated by an SLM based on the input text → additional guidance to the LLM.
- No additional knowledge resources required.
- Feedback ('Reward') can be used to retrain the SLM to generate hints to improve the direction given (so not necessarily better hints from a human perspective)

Input text
Article: (CNN) For the first time in eight years, a TV legend returned to doing what he does best. Contestants told to "come on down!" on the April 1 edition of "The Price Is Right" encountered not host Drew Carey but another familiar face in charge of the proceedings. Instead, there was Bob Barker, who hosted the TV game show for 35 years before stepping down in 2007. Looking sly at 91, Barker handled the first price-guessing game of the show, the classic "Lucky Seven," before turning hosting duties over to Carey, who finished up. Despite being away from the show for most of the past eight years, Barker didn't seem to miss a beat.

Reference
Bob Barker returned to host "The Price Is Right" on Wednesday . Barker, 91, had retired as host in 2007 .

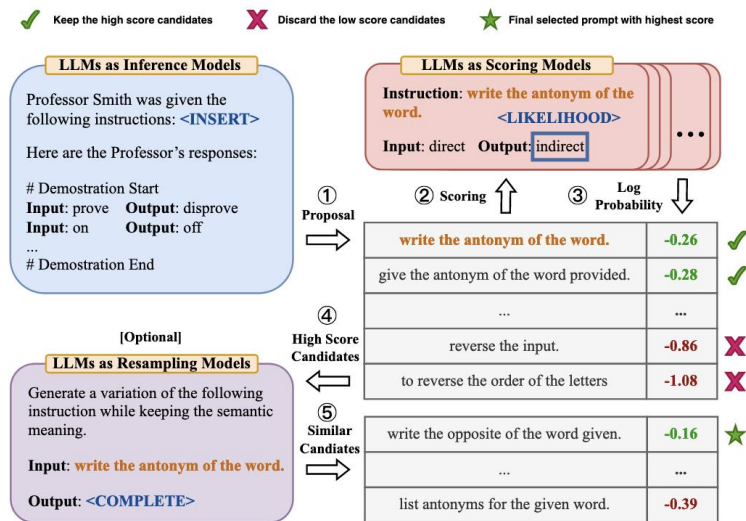
Directional Stimulus Prompting
Q: Summarize the above article briefly in 2-3 sentences **based on the hint**.
Hint: Bob Barker; TV; April 1; "The Price Is Right"; 2007; 91.
Model output

Standard Prompting
Q: Summarize the above article briefly in 2-3 sentences.
Model output
A: Bob Barker, the former host of "The Price Is Right", returned to the show for the first time in 8 years. He



Prompt Optimisation Techniques

- Prompts contain:
 - Instructions
 - Demos: input-output pairs that show how to solve a problem (e.g. CoT and ReAct)
- Given some value function and demos, prompt optimisation techniques aim to optimise value for instructions (i.e. prompt without demo).
- Automatic Prompt Engineering (APE; Zhou, et al. 2023) uses LLMs to optimise instructions using demos.



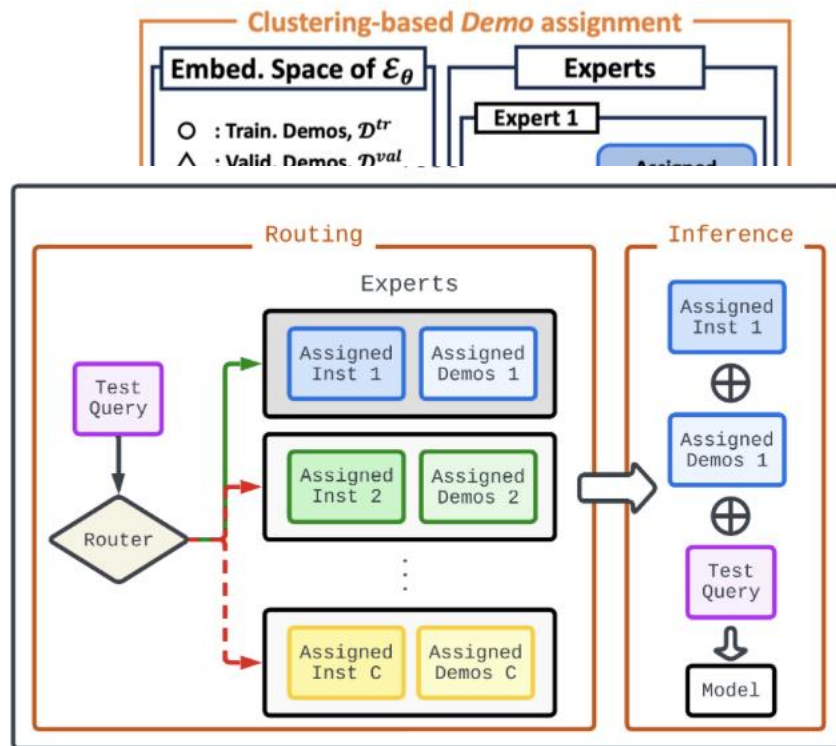
Zhou, et al. 2023

Mixture-of-Prompts (Wang *et al.*, 2024)

Mixture-of-Experts model applied to prompts construction.

✗ Most APE techniques use one model to generate only one demo-free instruction.

- Construction of prompts requires experts to focus on parts of the problem space $\rightarrow$ classify demos into clusters and assign them to an expert.
- The router decides which expert to assign to the query.

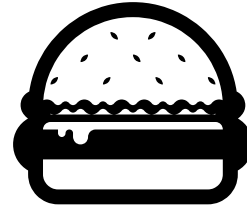


10-minute Break!

This will help you learn



10 minutes



What is the main advantage of Directional Stimulus Prompting over ReAct?



A. No additional knowledge required

B. No retraining required

C. No hallucinations

D. No prompt required

What is the role of experts in Mixture-of-Prompts?



- A. Experts are tasked with classifying demos into clusters
- B. Experts are used to generate specialised instructions and demos given an input query.
- C. Experts are solely responsible for evaluating demo-free instructions generated by a single APE technique.
- D. Experts are not involved in the Mixture-of-Prompts technique at all.

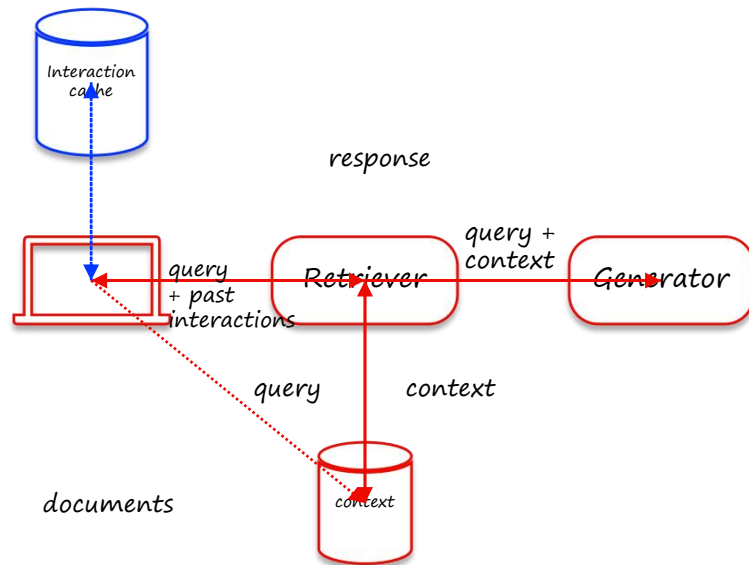
Retrieval Augmented Generation (RAG)



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RAGs with Memory

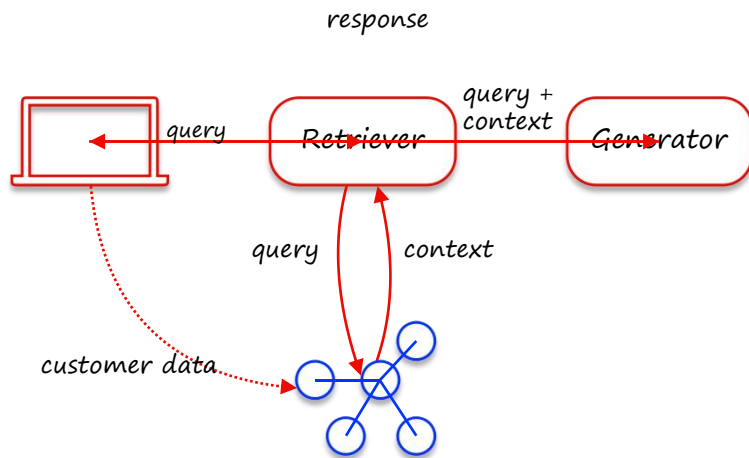
- A cache is used to store past interactions.
- Caches have an eviction policy:
 - Time-to-live (TTL): seconds until past interactions evicted.
 - Max memory: only a maximum amount of data is kept.
 - Least recently used (LRU): oldest interactions from a random sample are evicted. Requires `#samples` set.
 - Least frequently used (LFU): least used interactions from a random sample are evicted. Requires `#samples` set.
 - Random: random interactions are evicted.
 - None.
- Past interactions are passed on with the query



Graph RAGs

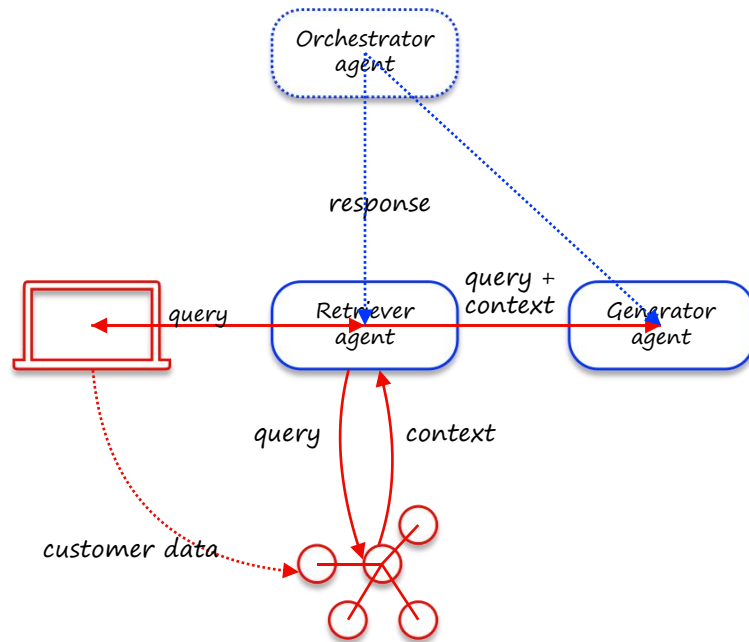
Used to store information about:

- Customers, to create detailed customer 360 views about their purchases, relationships, behaviours, etc..
- Domain processes/knowledge, also known as Knowledge graphs. Example applications include medical knowledge graphs.



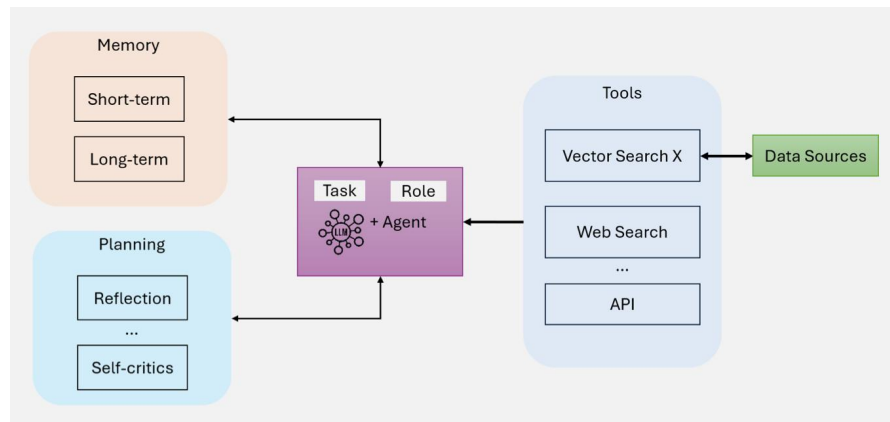
Agentic RAG (Singh *et al.*, 2025)

- Similar to microservices in software engineering, processes like the Retriever and Generator can be systems/components acting by themselves, i.e. agents.
- Agents can be orchestrated by another agent, i.e. orchestration, or they can act autonomously in a choreography.
- Agents are commonly used to act on behalf of the user, e.g. search the web



Components of an AI Agent (Singh *et al.*, 2025)

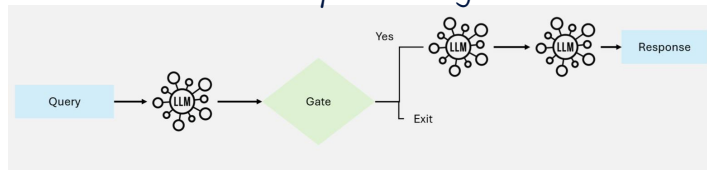
- LLM: primary reasoning engine and dialogue interface.
- Memory: stores short-term interactions and long-term knowledge.
- Planning: helps the agent reflect on, critique, and break down actions.
- Tools: capabilities that help the agent retrieve additional information



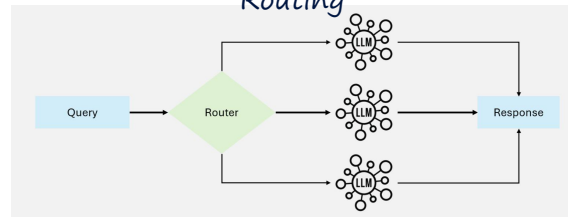
Singh *et al.*, 2025

Agentic Workflow Patterns (Singh *et al.*, 2025)

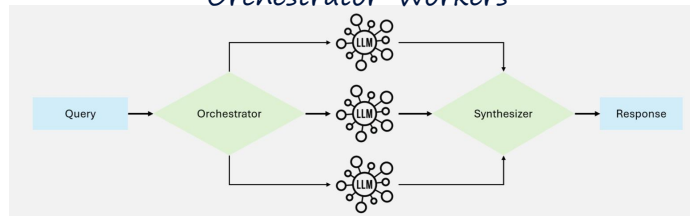
Prompt chaining



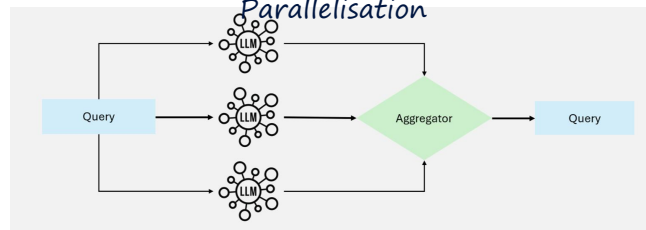
Routing



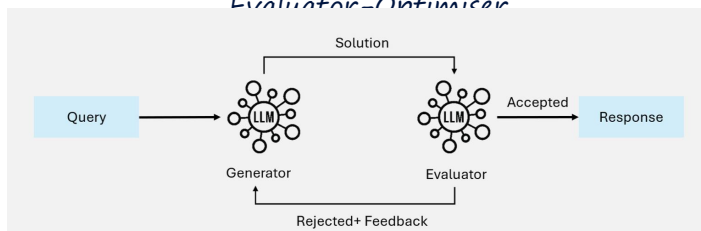
Orchestrator-Workers



Parallelisation



Evaluator-Optimizer



Challenges for Agentic RAG (Singh *et al.*, 2025)

- *Coordination Complexity: Managing interactions between agents requires sophisticated orchestration mechanisms.*
- *Computational Overhead: The use of multiple agents increases resource requirements for complex workflows.*
- *Scalability Limitations: While scalable, the dynamic nature of the system can strain computational resources for high query volumes.*



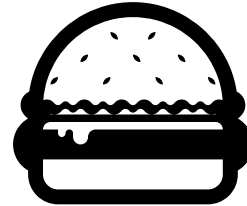
image of an unhappy police agent, MS Copilot

10-minute Break!

This will help you learn



10 minutes



Which is not a cache eviction policy in RAGs with Memory?



- A. Time-to-live
- B. Max memory
- C. Max processing
- D. Least frequently used



True or False

The Orchestrator-Workers Agentic pattern routes the query to one task-specific worker agent.

False

Fine-tuning



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Self-play fine-tuning (Chen *et al.*, 2024)

- *Similar to GANs:*
 - *Main Player (Discriminator): LLM that tries to distinguish opponent's response from human response.*
 - *Opponent (Generator): new version of the same LLM that tries to approximate/outperform human response ('ground truth').*
- Reinforcement Learning through Main Player's ranking of human vs. LLM response.
- Synthetic data is generated from original training instances to enable self-play.

Chosen: ground truth completion

...

In Southampton, what is the most popular form of transportation for commuters?

I don't have access to current data. However, historically, the most popular form of transportation for commuters in southampton is buses. The city also has a good network of trains and taxis.

Rejected: model generation @Iter 0

...

In Southampton, what is the most popular form of transportation for commuters?

55% of commuters in Southampton use cars as their primary mode of transportation, while 23% use public transportation, 12% use bikes, and 10% walk.

Model generation @Iter 1

I do not have access to the latest updates and statistics of southampton. However, based on previous reports, the most popular form of transportation for commuters in southampton is the public transport network, **which includes buses, trains, and ferries. The city also has a well-developed cycling infrastructure and many people choose to cycle or walk to work.** Additionally, **private cars** and taxis are also commonly used for commuting.

Chen *et al.*, 2024

Note: no human preferences needed!

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Mixture-of-Experts LoRA (MoLORA; Zadouri *et al.*, 2024)

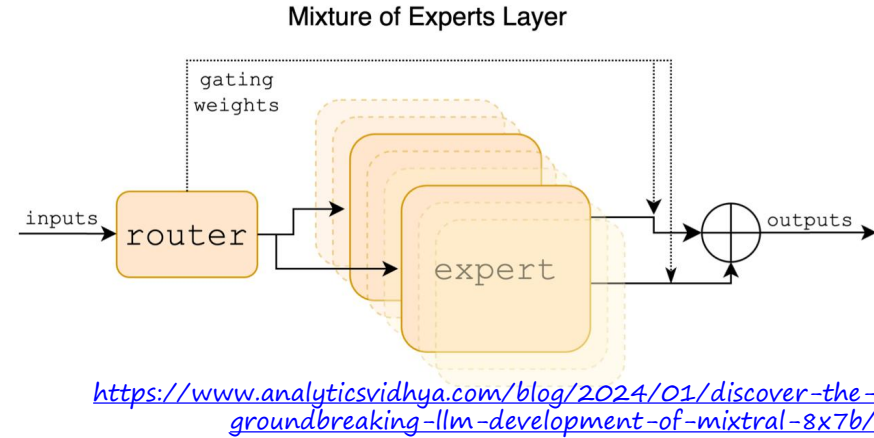
Multiple experts to determine parameter-efficient fine-tuning (i.e. matrices A and B).

🧑 10 experts lead to 3.18% trainable parameters vs 1% for traditional LoRA

✓ Outperforms traditional LoRA

✓ 'Generalises to unseen tasks'

Introduction of soft merging: weighted average of all experts in a routing block (instead of 1 or 2 experts).



Reinforcement Learning with AI Feedback (RLAIF)



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Disadvantages of RLHF

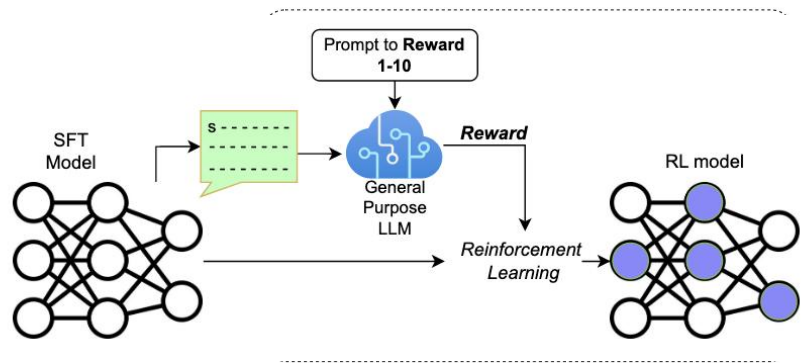
- ✗ *Time-consuming to collect human feedback*
- ✗ *Labour-intensive to collect human feedback*
- ✗ *Humans may be biased*
- ✗ *Humans may lack domain knowledge*



image of a defeated superhero that wears HHH on his chest, MS Copilot

Reinforcement Learning with AI Feedback (RLAIF)

- Replaces a human rater by an off-the-shelf LLM.
- Two types (Lee et al., 2024):
 - Canonical RLAIF: separate scoring and reward models
 - Direct RLAIF (d-RLAIF): only reward model



Lee et al., 2024

🧑 Equal performance to RLHF, but less time-consuming and labour-intensive.

Wrap-up



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Summary

- Trends in Prompt engineering include manual techniques like ReAct or Directional Stimulus Prompting, and automatic prompt engineering techniques such as Mixture-of-Prompts.
- Agentic RAG is the main trend in Retrieval Augmented Generation.
- New developments in fine-tuning include Self-play fine-tuning, Mixture-of-Experts LoRA, and RLAIIF.

For next lecture

 *Assignment 1 - Prompt engineering is due this week!*

No reading material for next week.

Take Home Message...

*Consider the message that you want to create when
designing your interfaces.*



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Things to think about

What is the purpose of prototyping?

*What is the difference between Lo-fi
and Hi-fi designs?*

*What is a storyboard? When can
these be used?*

What should you show your client?

References and Resources

- Chen, Z., Deng, Y., Yuan, H., Ji, K., & Gu, Q. (2024). *Self-Play Fine-Tuning Converts Weak Language Models to Strong Language Models* (No. arXiv:2401.01335). arXiv. <https://doi.org/10.48550/arXiv.2401.01335>
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- Zhou, Y., Muresanu, A. I., Han, Z., Paster, K., Pitis, S., Chan, H., & Ba, J. (2023). *Large Language Models Are Human-Level Prompt Engineers* (No. arXiv:2211.01910). arXiv. <https://doi.org/10.48550/arXiv.2211.01910>

Questions



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